

THE OUTLIERS

Mumbai Rent Radar

What should a flat *actually* cost?

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The same 2BHK, ten different prices

- ▶ **The question:** every Mumbai tenant asks what a flat should really cost.
- ▶ **The noise:** identical flats are advertised at wildly different rents.
- ▶ **Our job:** build a fair-rent model and flag listings that look mispriced.

THE ANSWER, IN ONE LINE

Rent is set by **how big** a flat is —
far more than **where** it is.

We predict a fair rent within ~27% for ordinary flats and flag the mispriced ones.

From messy listings to a table we can trust

- ▶ **5,100** → **5,050 listings** after removing duplicates.
- ▶ **6 fields** collapsed from dozens of spellings to real categories.
- ▶ **Prices unlocked** from text like ₹1,47,500, 50k and 40,000/-
- ▶ **98% join** to a locality attributes table.

5,050

clean listings

98%

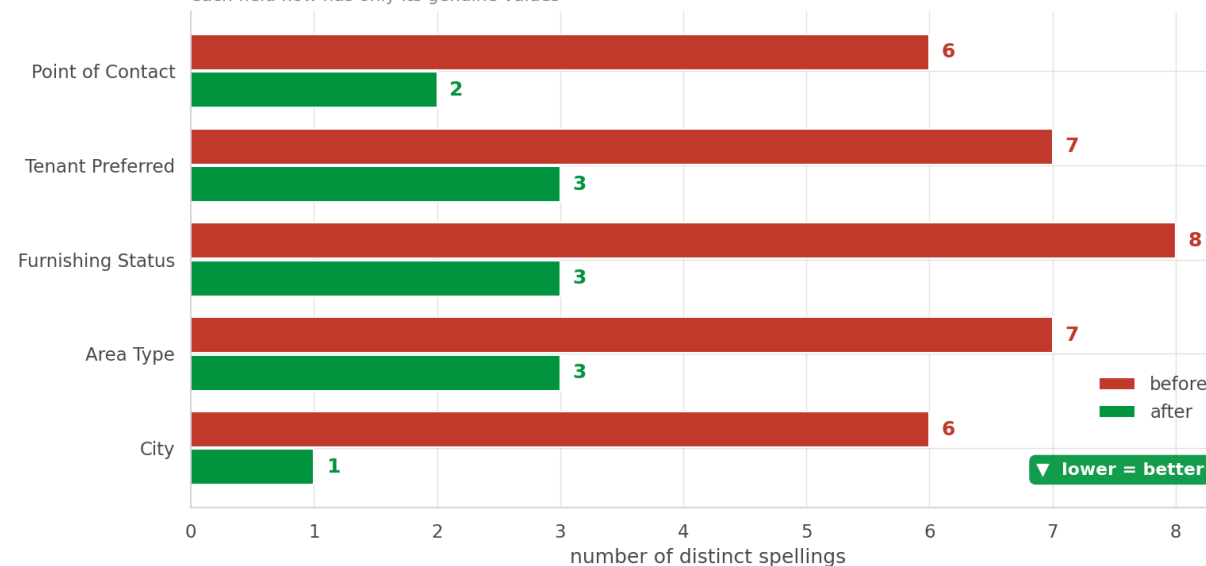
locality match

10.8%

gaps imputed

Cleaning collapsed duplicate spellings into real categories

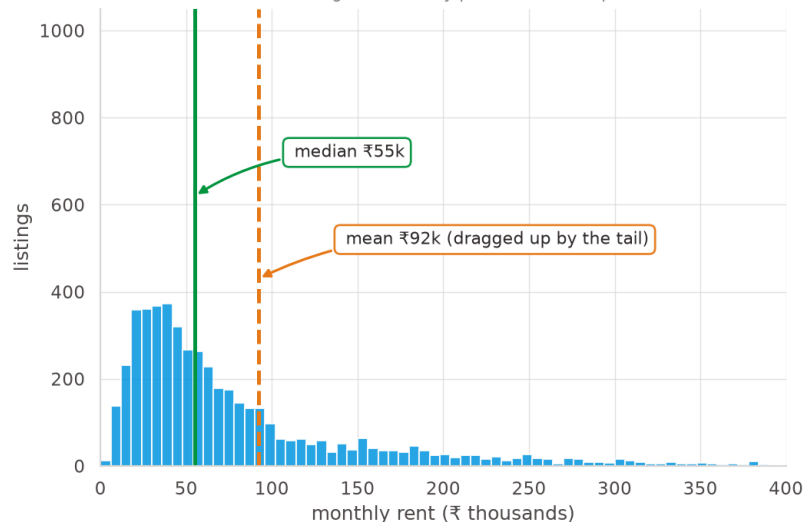
each field now has only its genuine values



Rent is skewed — most flats are affordable

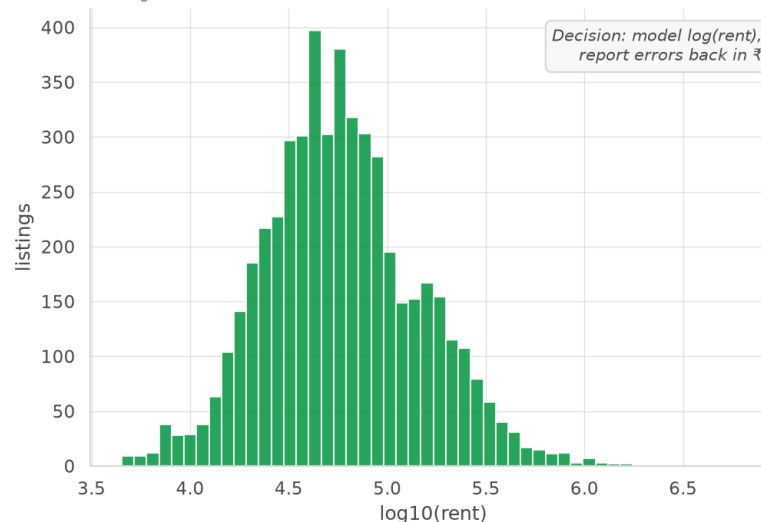
Rent is heavily right-skewed

most flats cluster low; a long tail of luxury pulls the mean up



...but $\log(\text{rent})$ is almost symmetric

the log scale tames the tail — friendlier for a linear model

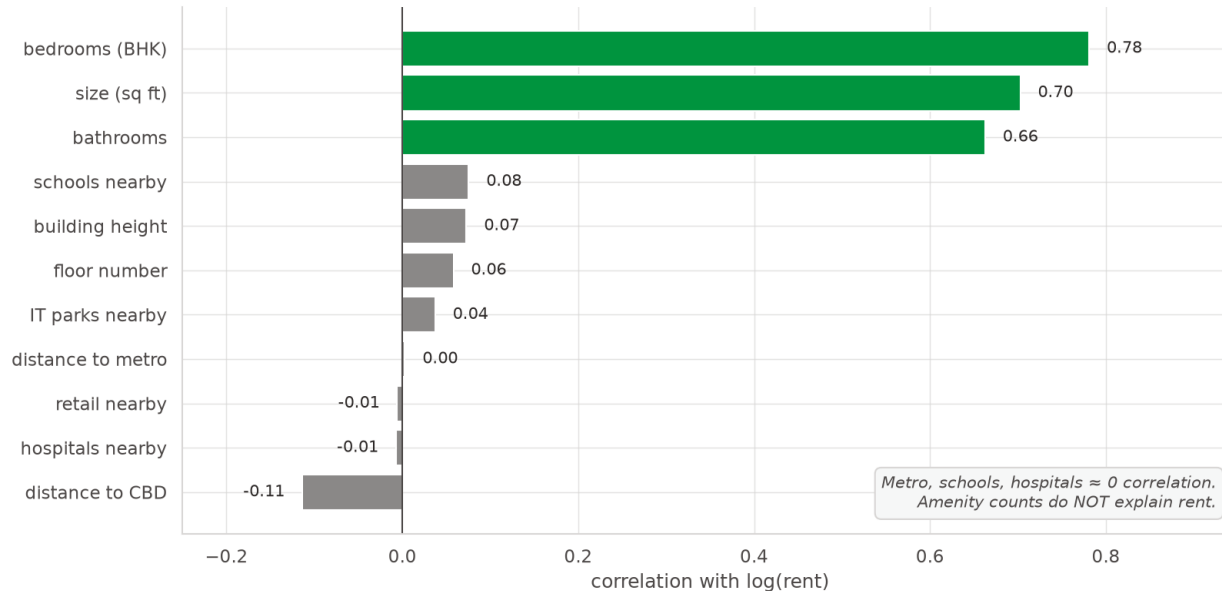


- ▶ **Median ₹55k** but the mean is ₹92k — a long luxury tail.
- ▶ **We model $\log(\text{rent})$** so a ₹10k miss is judged fairly at every price.
- ▶ **Errors reported** back in rupees for the decision owner.

In a location-obsessed city, location barely moves rent

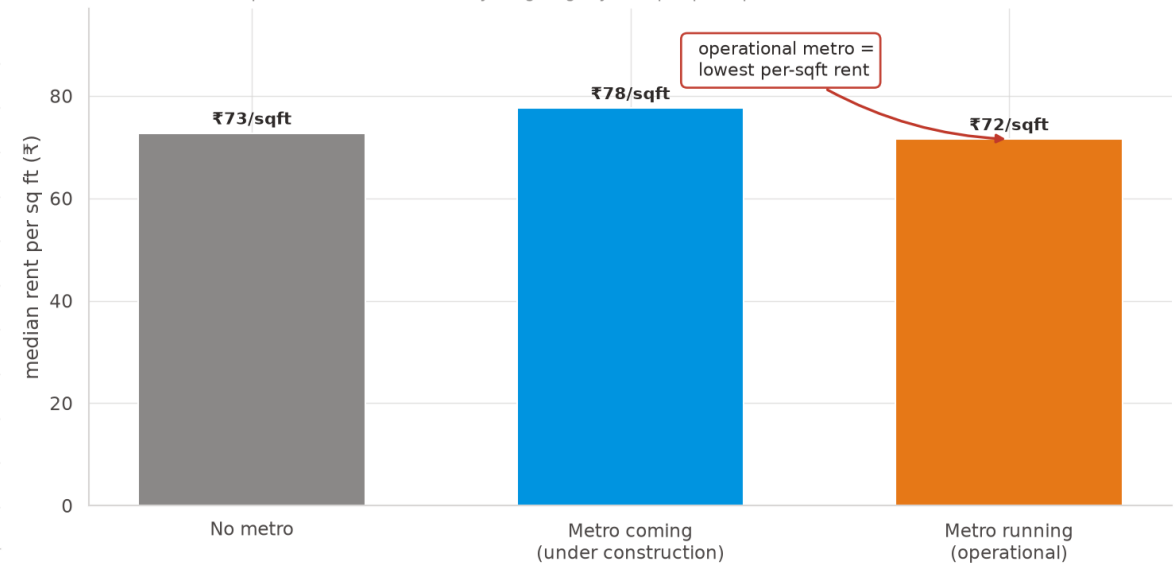
The flat sells itself — the neighbourhood barely matters

size, bedrooms and bathrooms tower over every location amenity



A running metro does NOT raise rent in Mumbai

localities with an operational metro are, if anything, slightly cheaper per sq ft

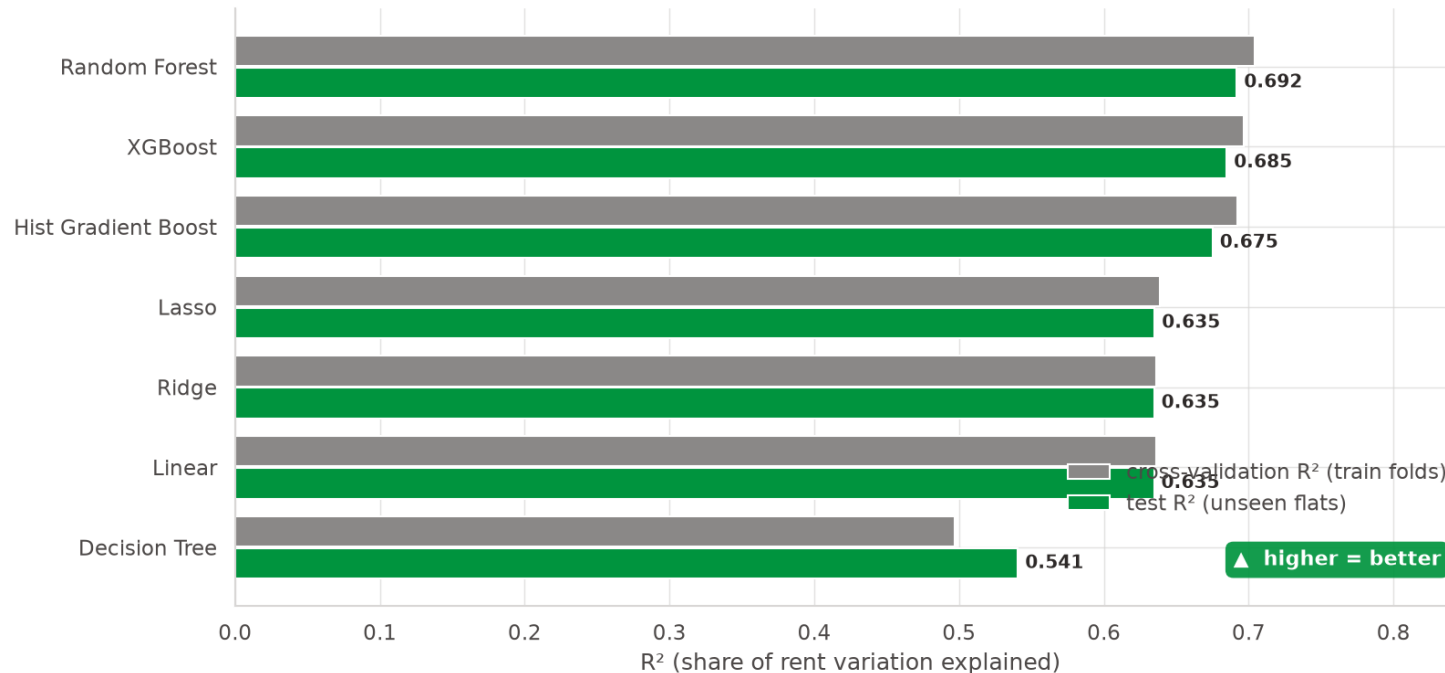


Size, bedrooms & bathrooms carry the price; metro/schools ≈ 0 . A running metro is even slightly cheaper — verified, not a bug.

We benchmarked seven, and let the best one win

Random Forest explains the most rent — and it generalises

cross-validation $R^2 \approx$ test R^2 : the score holds up on flats the model never saw



- ▶ **Random Forest wins** on unseen flats and on rupee error.
- ▶ **It generalises** cross-validation $R^2 \approx$ test R^2 .
- ▶ **One final model** chosen honestly from the field.

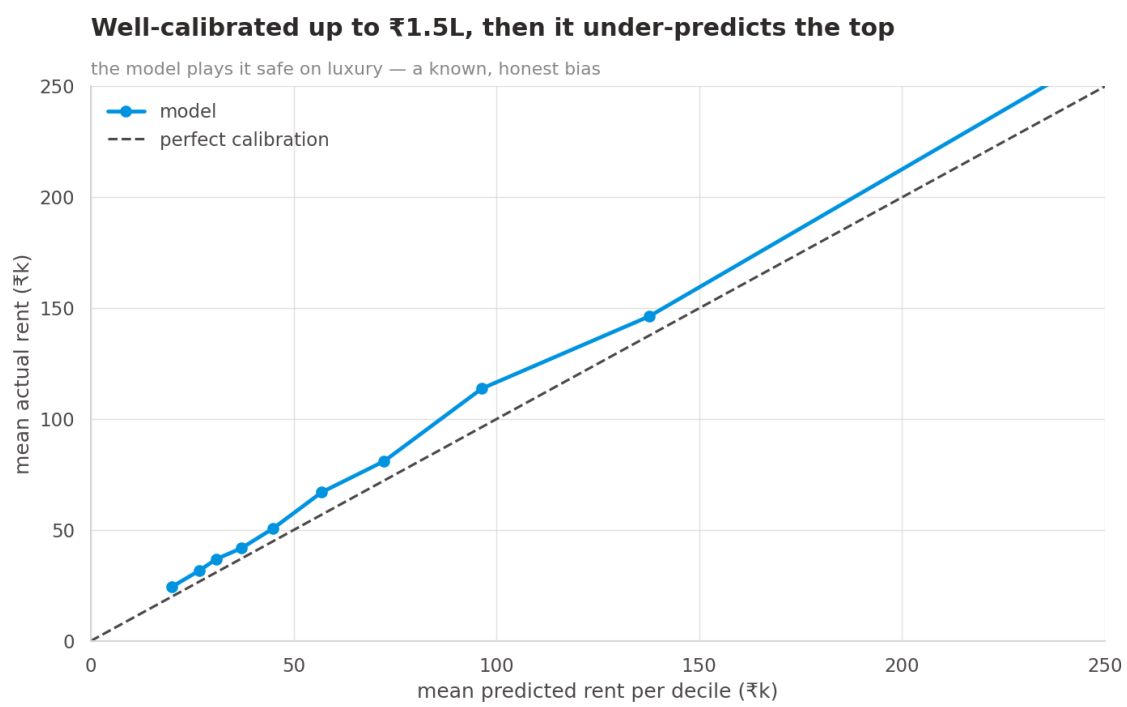
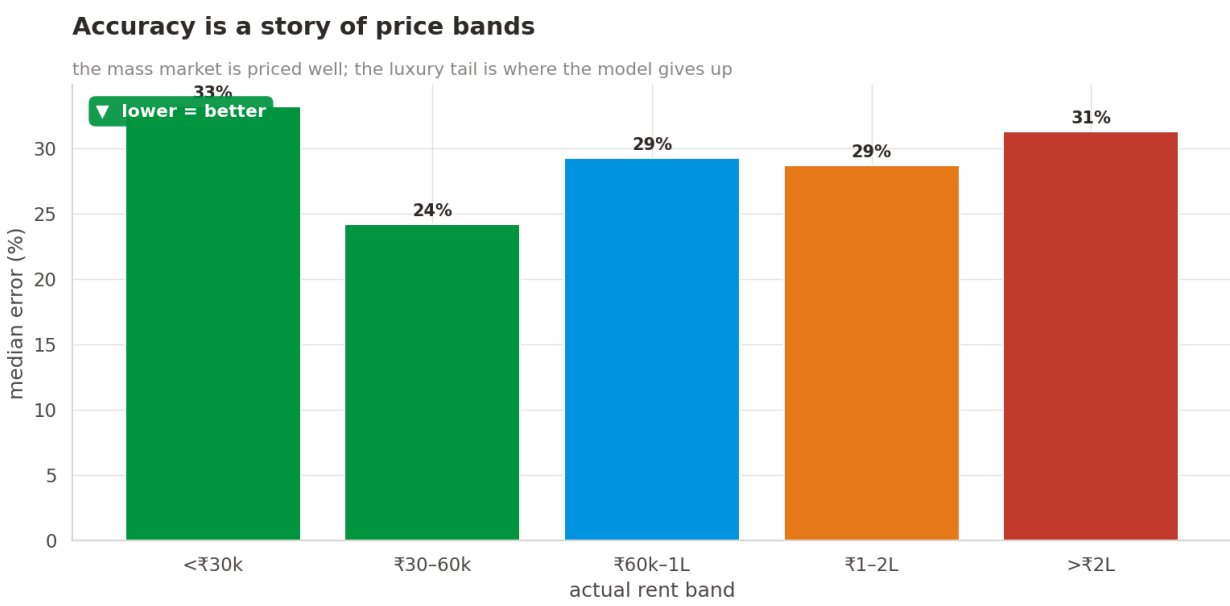
0.70

R^2 (unseen)

27%

typical error

Reliable for the mass market, loose for luxury



Under ₹1L the model is dependable; above ₹2L it under-predicts a thin, unique tail — a known, safe bias.

A mispriced-listing radar for renters

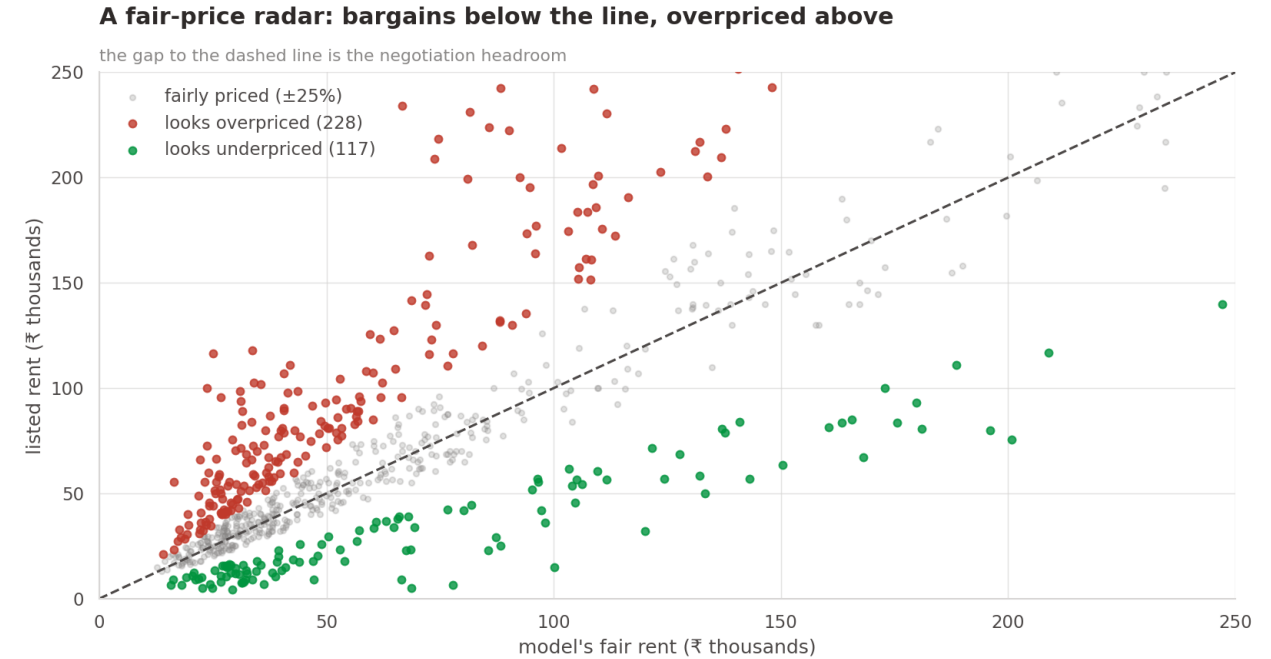
- ▶ **The gap is the signal** listed rent vs the model's fair rent.
- ▶ **Bargains below the line** overpriced above it.
- ▶ **Ready to ship** the fastest win for a pricing team.

221

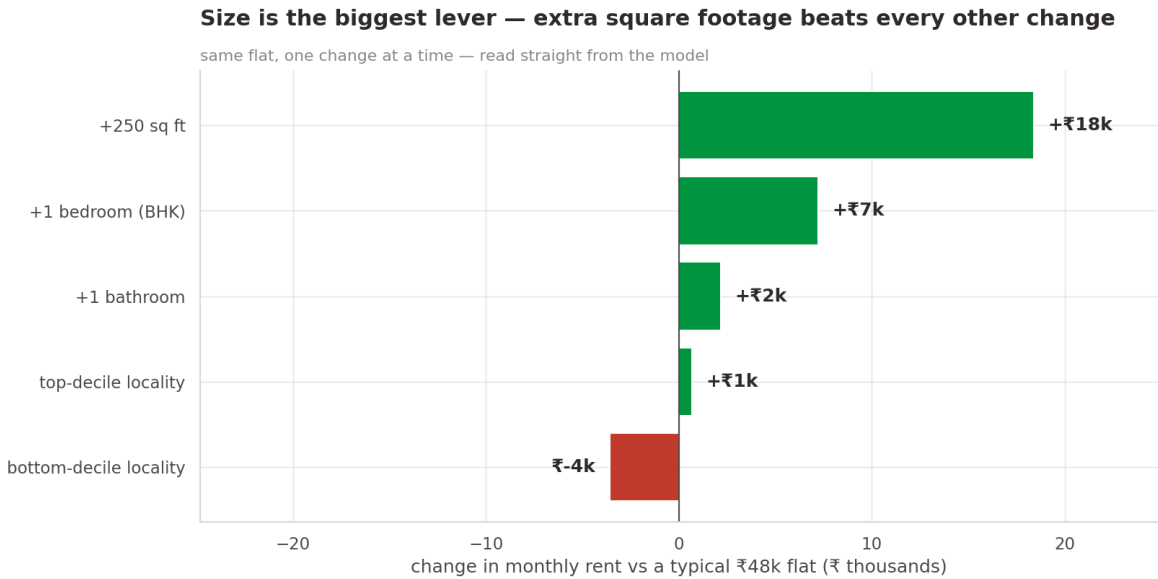
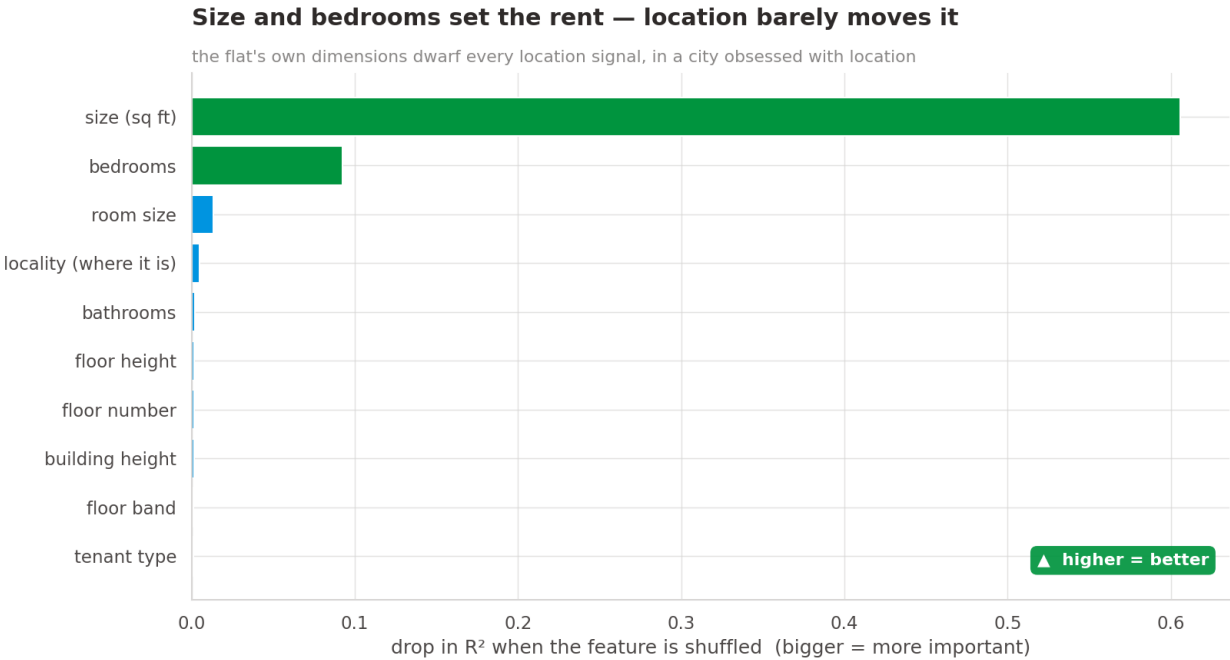
flagged overpriced

117

bargains found



Size first — and what an upgrade is worth



Adding 250 sq ft beats an extra bedroom, a bathroom, or a jump to a top-tier locality. Location sets the baseline; size sets the price.

What NoBroker's pricing team should do

- ▶ **Ship for the mass market (<₹1L)** — reliable, and where most renters are.
- ▶ **Launch the mispriced detector** — 338 flags on the test set alone.
- ▶ **Price by size first** — don't pay up for metro/amenity flags.
- ▶ **Refresh the data** — it's one snapshot; watch for drift.
- ▶ **Better features, not more rows** — the learning curve is flat.

THE AI MOMENT WE CAUGHT

An early model scored $R^2 > 0.9$.

That was the tell — we traced it to a data leak and removed it.

AI drafts fast; we verify before shipping.

Bottom line

Mumbai rent is about how big, not just where. Ship the estimate, flag the mispriced, price by size.